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Nature-inspired in-flight foldable rotorcraft

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Supplementary material for this article is available [online](#)

Abstract

The paper presents a novel rotary wing platform, that is capable of folding and expanding its wings during flight. Our source of inspiration came from birds' ability to fold their wings to navigate through small spaces and dive. The design of the rotorcraft is based on the monocopter platform, which is inspired by the flight of Samara seeds. The wings are constructed by applying origami techniques to fold them in flight. Two configurations are presented, featuring active or passive mechanisms for wing-folding depending on specific application requirements. The two configurations can reduce their overall footprint by approximately 39% and 69% while in flight. A cyclic controller is implemented for controlling the translational motion, where the direction is controlled by pulsing the motors at a specific instance during each cycle of rotation. We have presented experimental results to prove the control of our platform in different modes while in flight. The presented platforms enhance the practical uses of the monocopter platform by providing it with the ability to reduce its footprint while in flight actively, or by allowing them to dive through the air without any additional actuator.

1. Introduction

In recent years, unmanned aerial vehicles (UAVs), in particular quadrotors, have become an integral part of many industries [1]. Quadrotors have proven to be incredibly useful for various applications such as delivering packages, inspecting building structures, surveillance, etc [2]. Their maneuverability is a significant advantage as it allows them to safely and efficiently access hard-to-reach or dangerous locations, providing an alternative to human labor. However, due to aerodynamic lift being generated directly by the propellers, quadcopters are not the most power-efficient vehicles. This is because according to the momentum theory, the hovering power is inversely proportional to the rotor radius [3]. To solve this problem, efforts have been made, such as [4–6]. In contrast to those traditional multi-rotors, fixed-wing aerial robots show much higher power efficiency by making advantage of their huge airfoils, but these are not suitable for operations in confined spaces due to their forward velocity, as well as they lack the ability to hover and vertical take-off and landing. Similar to the

fixed wings, rotating aerial vehicles have also shown high flight efficiency [6–8].

Monocothers are a class of samara-inspired rotorcrafts, which, unlike quadrotors, rely on the lift being generated by their entire wing. They fly by constantly rotating about their yaw axis, using a single wing to generate the lift required for hovering. Recent research has proven that this configuration is far more efficient compared to the quadrotors [7]. Samara seeds are known for their ability to autorotate while falling to slow their descent rate and disperse. Monocothers carry the inherent advantage of autorotation from samara seeds, which helps them to descend gracefully in case of a power failure. The concept of monocothers is not new and has been researched in different studies. Although the first implementation of this theory dates to the 1900s, the recent developments began with the publication by MIT in 2008 [9].

Some of the applications of monocothers include unpowered lightweight sensor [10], short-range urban surveillance [11], LiDAR inertial odometry [12], and passively scanning and mapping the

surrounding environment using simultaneous localization and mapping [13]. Monocopters usually have two actuators to achieve flight and control: a motor and a servo to control the flap. The concept of single actuator monocopter was first introduced in [14], where the authors utilized a single motor for directional as well as altitude control of monocopter. On the other hand, a recent study [15] demonstrated the use of monocopters in a two-flight mode capable rotorcraft, where the other flight mode was obtained by implementing extra actuators for the mode. The monocopters concept has been extended to dual wings for the development of transformable hovering rotorcraft in [16], where the authors have utilized four actuators to control the altitude and direction of the UAV. The dual-wing configuration was also used as a cooperative configuration in [17], where the authors presented the capability of this platform to fly in a dual-wing configuration and passively separate during flight. In [8], the authors have presented a modular version of the monocopter platform, whereby they have demonstrated the adaptability and control of this platform using different configurations.

Traditionally, the monocopters have been developed using lightweight materials such as carbon fiber [18], foam [9], balsa wood [19] or a combination of balsa wood and foam [14]. This ensures that the wing remains light as well as rigid, to provide lift when flying at a certain angle with respect to the oncoming airflow. The first attempt to use a flexible, semi-rigid wing for a monocopter was done in [20], where the authors introduced a foldable single actuator monocopter that can be folded up into a compact pocketable form for storage when not in flight. However, to the best of the author's information, the concept of foldable wings for rotorcrafts during flight has not been researched. In this work, we aim to implement the concept of foldability on monocopters during flight. Through active and passive control, we aim to control the wingspan of the aerial vehicle during its flight.

Several aerial vehicles capable of actively changing shape have been previously developed [21–24]. Self-foldability and self-deployability are explored in [22]. Active morphology of the shape of the robot helped achieve multi-modal flying and walking ability in [21], whereas, in [23], active morphology contributed to negotiation through narrow gaps, close inspection of surfaces, and object grasping. In [24], active control over the folding of wings helps improve the robot's maneuverability, agility, and stability.

Several designs have also been proposed that enable aerial vehicles to passively change the shape of their body [25–29]. Bouman *et al* [25] and [26] demonstrated the automatic unfolding of an aerial vehicle after launch, while [27] presented a manually foldable wing drone with rapid deployment

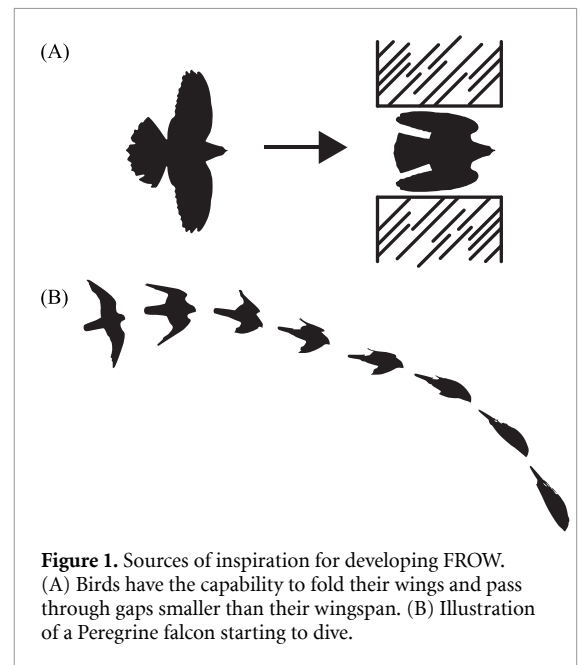


Figure 1. Sources of inspiration for developing FROW. (A) Birds have the capability to fold their wings and pass through gaps smaller than their wingspan. (B) Illustration of a Peregrine falcon starting to dive.

capabilities. Although these aerial vehicles could be ideal for compact storage and rapid deployment, they do not focus on repeatedly changing the shape after deployment and therefore require manual assistance to be returned to their compressed forms. Bai *et al* [28] features passive variable-sweep wings for effective pitch maneuverability without active control surfaces, whereas, [29] presented a quadcopter that can modify its shape mid-flight by adjusting thrust forces.

We present the design and development of the foldable rotary origami wing (FROW). One of the drawbacks of monocopters during their flight is their large footprint which hinders their ability to maneuver through tight spaces. FROW draws its motivation from the flight of birds. These feathered creatures possess the ability to fold their wings elegantly when navigating through narrow gaps with ease (figure 1(A)). This remarkable phenomenon is what FROW strives to capture. FROW can fold its wings during flight to reduce its footprint, making this platform more flexible in its usability. We introduce two different configurations of this platform which have similarities in their principle and design concept; however, they differ in their folding mechanism and practical applications. FROW-A can actively control its wing folding mechanism using actuators, whereas FROW-P's wing folding is passive and requires no actuators.

FROW-A is designed to actively control its wingspan in flight, allowing for alterations in its footprint. This feature enables the extended wing configuration of FROW-A to be utilized for its power-efficient flight, while the folded wing configuration facilitates maneuvering through narrow spaces and obstacles when required. Reducing the footprint in flight can also help in obtaining a stealthy approach to avoid radar

detection as well as aggressive maneuvers. In addition, modifying the wingspan also affects the rotational speed of the aerial vehicle. Consequently, active control of rotational speed eliminates the need for a camera or Lidar sensor tuning and allows direct synchronization with optimal rates required while airborne.

In comparison to the foldable monocopter developed by Win *et al* in [20], FROW-P boasts a unique feature where it eliminates the need for manual assistance for both folding and unfolding. The novel configuration permits the wing to automatically extend during takeoff and retract with ease when approaching landing. Sometimes, birds fold up their wings behind their body to quickly plunge through the air. Cape Gannet uses this ability to take nose dives toward the water to dive deeper, whereas Peregrine Falcon uses the same principle to simply hurtle thousands of feet through the air at high speeds to quickly descend toward its prey (figure 1(B)). FROW-P is designed to mimic the diving of these natural fliers. A previous work done by Win *et al* [30] achieved the diving in the monocopter by utilizing its flap. By deflecting the flap to a large angle of attack, the authors managed to deflect the airflow to enter a dive mode. However, FROW-P does not require an additional actuator to dive. With the help of its passive wing folding mechanism, FROW-P can plunge through the air, which can help it in diving and quickly descending. This type of maneuver can be ideal for bypassing harsh weather conditions when flying at high altitudes. The maneuver can also be employed for a more aerobatic and agile monocopter depending on the application. Changing the folding mechanism parameters allows FROW-P to be flown with different wing spans. Additionally, the same parameters can be changed to fly at a desired rotational speed as well.

This work will make the monocopter platform more versatile in its field. The novelty of this research and its contributions can be summarized as follow:

- The design and development of two new monocopter-based prototypes, named FROW-A and FROW-P, are presented, which can fold their wings based on active and passive control mechanisms during flight. The resulting configurations can reduce their overall length by 39.18% and 65.19% respectively, while still in the air.
- Experimental verification is performed to observe the FROW's performance during the folding and unfolding of wings during flight. We have presented results for position hold during closed-loop waypoint and trajectory tracking experiments. Experimental results for expanded and folded modes while holding the position are presented, verifying the practical use by reducing the footprint and maneuvering through small gaps.

- Experimental proof for the passively foldable mechanism has been presented, where we have demonstrated successful flights using the platform featuring its diving and recovery capability in flight.

In the following section, we elaborate on the design principles of FROW. The section also describes the different parts and components onboard the FROW prototypes. Section 3 describes the dynamic modeling and flight control of the platform. Section 4 discusses the results followed by a conclusion in the final section.

2. Prototype design

In this section, we describe the design and manufacturing of the wing and onboard electronic components of the aerial vehicle. The contraction and expansion mechanism, its modeling, and verification of the modeling using experimental techniques are discussed for the two different prototypes.

2.1. Origami-inspired wing

Origami is the traditional Japanese art of paper folding. The core technique that lays the foundation of origami is folding. The two most common types of folds used for any origami model are mountain folds and valley folds, which have been utilized in the design process of FROW's wing. The wing of FROW is designed in such a way that it allows an accordion-style folding. This allows the wing to obtain a reduction in length in the desired direction. The panels fold downwards, allowing FROW to fold its wing and reduce its footprint. The downwards fold also allows the carbon fiber rods to squeeze in without being blocked by the wing panels in the FROW-A prototype. In a traditional accordion fold mechanism, the hanger supports are connected to the middle of the panels using a swivel mechanism. We modified the mechanism by attaching the supports to the folds. This helps in simplifying the mechanism by removing the need for a swivel structure for the supports. The final design is able to encapsulate the features of a monocopter wing, which is required to be rigid in flight, as well as fold as desired. Figure 2 depicts the FROW-A prototype during its flight in extended and folded wing configurations.

The wing is designed to keep sufficient gaps in between for the fold and spare enough room for attaching the hanger supports. The hanger supports slide on the carbon fiber rod, supporting the wing and folding mechanism span-wise. The carbon fiber rod acts as the guide track for the wing panels. The shape of the wing is chosen to mimic that of a samara seed as closely as possible to inherit the natural properties of the seed.

The material selection for the wing was done carefully to attain a level of rigidity sufficient for

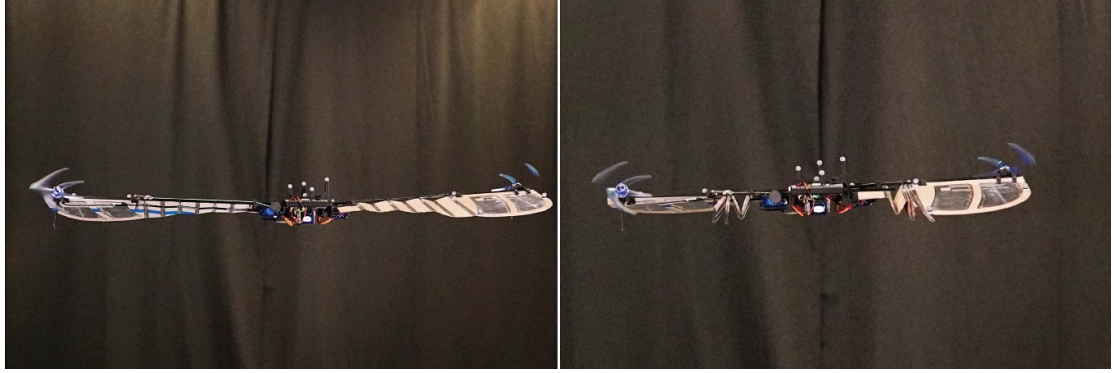


Figure 2. An image of FROW-A in flight in the extended and folded wing configurations.

producing advantageous aerodynamic forces when expanded, while simultaneously possessing adequate flexibility to facilitate folding. The balsa wood panels (1 mm thickness) are meticulously cut out and laminated using a thin plastic film. The folded-up wing is then subjected to hot air (150 °C) which helps in retaining the fold shape and permits the desired wing-folding capability. The hanger supports are then installed at appropriate distances and the finished wing can be mounted on the carbon fiber rod.

FROW-A is designed such that the wings only fold by approx. 40% of the total wingspan, leaving the remaining wingspan in the folded mode. This design restriction is necessary to ensure adequate wing area remains available for lift generation while the wings are folded. During preliminary experimentation, it was observed that using a single wing caused FROW-A's rotational speed to be extremely high and thereby placed significant stress on both the motor and servo components. To overcome this challenge and alleviate pressure on actuators, two wings were incorporated into FROW-A instead of one to maximize lift production in the folded mode. Both wings overlap the carbon fiber rod protruding out on folding which ensures optimal use of available space.

FROW-P is designed to obtain a reduction in the wing surface area passively, through the application of a spiral spring. The wing folds by approx. 70% of the total wingspan. The remaining wingspan is mostly the folded sections of the wing, which is unable to generate lift for the monocopter flight, therefore, putting FROW-P in dive mode. As observed in the FROW-A prototype, the dual wings overlap the carbon fiber rods upon folding. Therefore, to eliminate any extra portions of carbon fiber rod protruding out, we designed a telescopic rod as depicted in figure 3. The telescopic mechanism is mechanically rigid enough to eliminate any twisting of the wing, whilst being smooth in its motion.

Table 1 provides a comparison of the wing parameters of the two prototypes in their expanded and folded configurations, comparing the effective span length, footprint, and overall footprint percentages.

2.1.1. Contraction and expansion of panels

On the passively controlled FROW-P, the folded wing is held in tension using a spiral spring. Figure 4 shows an illustration of the unwinded and winded spring model for the FROW. Some assumptions are considered for approximating the spring force formula [31], (1) the cross-section of the spring is rectangular and its thickness is even throughout its length, (2) the string used for the mechanism is not elastic and the angle of deflection obtained is linearly proportional to its displacement. Therefore, the spring force F_s is approximated as,

$$F_s = K_s \Delta x + F_{s0} \quad (1)$$

where K_s is the spring constant, Δx is the displacement of the string, and F_{s0} is the pre-loaded spring force.

The centrifugal force F_c acting on FROW-P can be estimated by approximately dividing the UAV into two parts about the center of rotation, as shown in figure 4,

$$F_c = m_1 \Omega_z^2 r_1 = m_2 \Omega_z^2 r_2 \quad (2)$$

where m_1 and m_2 are lumped masses located at distances r_1 and r_2 from the center of mass of the UAV, respectively. Here, we can consider an assumption that the mass m_1 depicted in figure 4 is identical for both unwinded and winded springs. This is because the distance r_1 of mass m_1 from the center of mass does not change significantly in the two cases.

The balance of forces in the folding mechanism determines the condition of flight for FROW-P as,

$$\begin{aligned} \text{To fly} \quad & F_c > F_s \\ \text{To fold} \quad & F_c < F_s. \end{aligned} \quad (3)$$

Once the rotorcraft starts to spin, the centrifugal force applied by the motor and its support structure expands the wing outwards. The mechanism stores

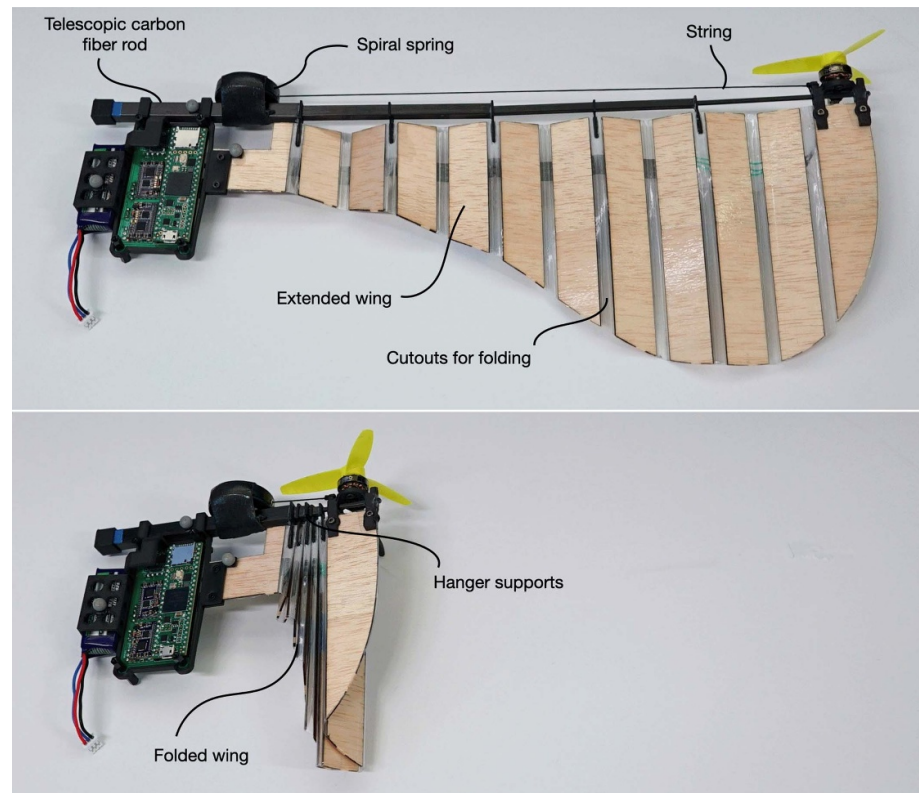
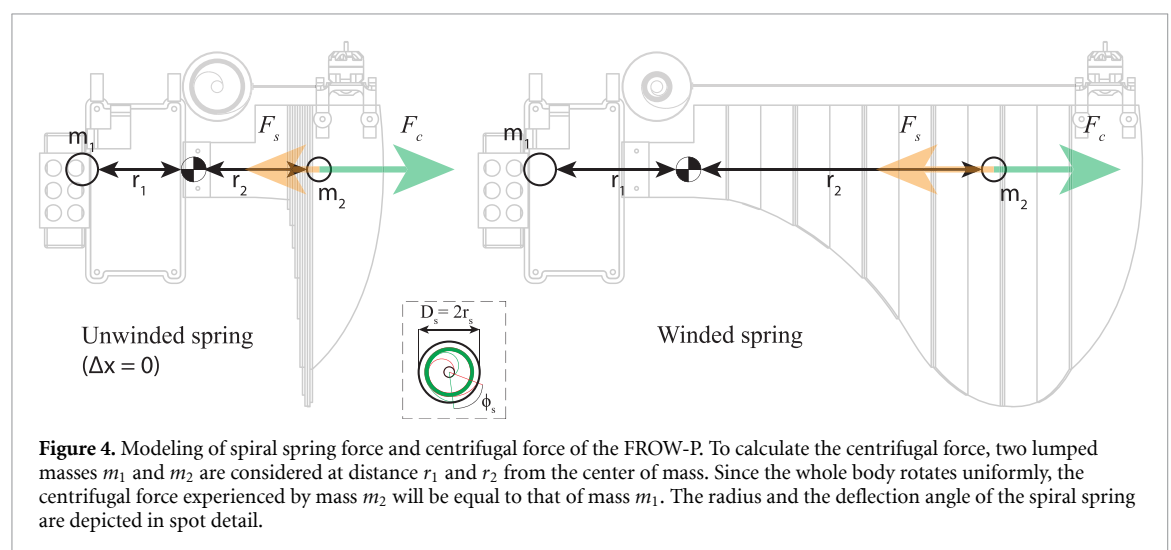


Figure 3. A prototype of FROW-P showing the extended and folded wing in the top and bottom images, respectively.

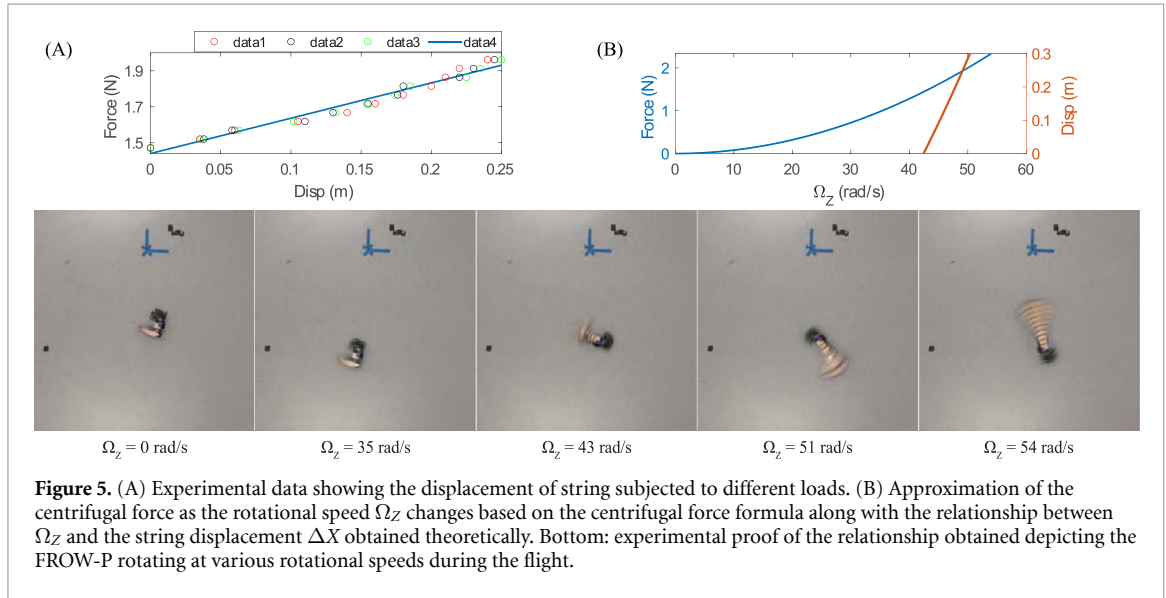
Table 1. Comparison of expanded and folded configurations.

Prototype	Status	Effective length (mm)	Features footprint (mm ²)	Footprint (%)
FROW-A	Expanded	735	2.3091×10^3	100
	Folded	447	1.4043×10^3	60.8
FROW-P	Expanded	385	1.0053×10^3	100
	Folded	134	314.1593	31.25



energy in the spiral spring which folds the wing as soon as the effect of centrifugal force is not experienced by the motor.

The constant K_s of the spiral spring is approximated by experimentation where a set of weights are used to measure the extension in the length of



the string under different loading conditions. The mechanical properties of different, readily available springs were tested for the selection of an appropriate spiral spring, which can provide the desired results. Figure 5 depicts the results of the experimentation performed, showing the results from three different trials. The regression line shows that there is nearly a linear relationship between the force applied and the displacement of the string. The figure also depicts the relation between the rotation speed Ω_Z and the centrifugal force, approximated using equation (2).

Considering a hypothetical equilibrium, where the forces F_c and F_s are equal, a relation between the rotation speed Ω_Z and the displacement of the string can be determined as,

$$m\Omega_Z^2 r = K_s \Delta x + F_{s0} \quad (4a)$$

$$\Omega_Z = \sqrt{\frac{K_s \Delta x + F_{s0}}{mr}}. \quad (4b)$$

Plot(C) in figure 5 presents the theoretical relationship between Ω_Z and ΔX . We conducted experiments using the FROW-P prototype for verification of this relationship. As depicted in the frames of the flight video in figure 5, the experimental results followed the trend closely. It can be observed that the centrifugal force in FROW-P overcomes the spring force at around 43 rad s^{-1} and achieves full extension at 54 rad s^{-1} .

On the other hand, FROW-A employs a servo motor to fold the wing. The servo is connected to a spool, which reels in the string connected to the other side of the wing, which folds the panels. For the expansion of the wing, the servo relaxes the string, and the rotating wing expands under the influence of the centrifugal force experienced by the wing, motor, and support structure attached to it. The length of the

wing to be folded in was carefully chosen based on experimentation, leaving a sufficient area of the wing for the folded model to fly with. Figure 6 depicts the folding principle employed on the FROW-A.

2.2. Electronics

The FROW is based on a custom-built flight controller system, which allows it to be controlled by a human operator or through a closed-loop feedback position controller on a PC in a motion-capture environment. The flight module is based on a Teensy micro-controller which has an inbuilt micro SD card reader for data-logging capability. For heading reference, a high-performance three-axis magnetometer is used on board for control in an open loop. All electronic components including the flight controller, receivers, magnetometer, power regulator, and the electronic speed controllers (ESCs) are arranged and soldered on a custom-designed printed circuit board, which is mounted on a 3D-printed central hub of the FROW. This hub also houses the Li-Po battery which powers the FROW.

3. Dynamic modeling and control

In this section, we discuss the dynamic modeling of the prototype and the controller applied to obtain altitude and transitional control of FROW.

3.1. Dynamic modeling

As depicted in figure 7, a right-handed inertial frame is defined using $(X, Y, Z) \in I$ and the body frame attached to the center of gravity is defined using $(x, y, z) \in B$. Using the Newton–Euler formulation, the translational dynamics of the UAV can be expressed as,

$$m\ddot{\mathbf{p}}^I = \mathbf{R}_B^I \mathbf{F}^B + m\mathbf{G} \quad (5)$$

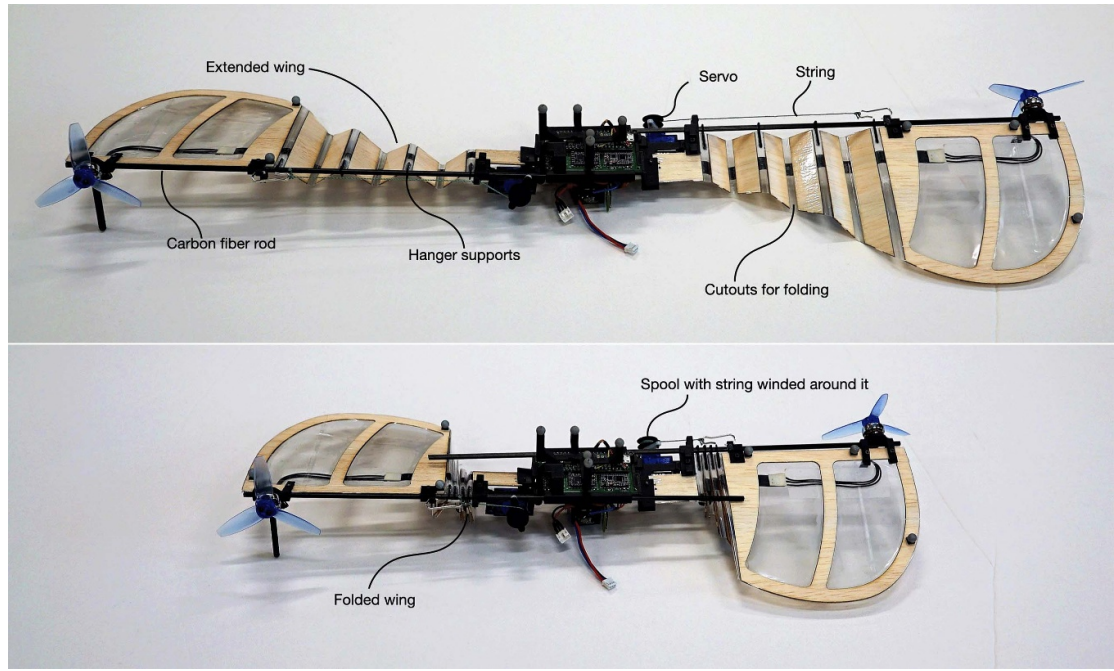


Figure 6. A prototype of FROW-A showing the extended and folded wing in the top and bottom images, respectively.

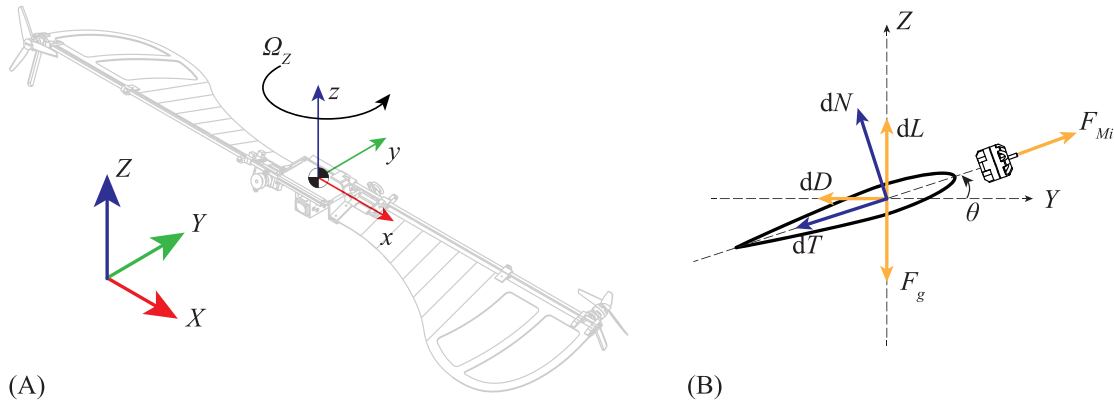


Figure 7. (A) This figure describes the inertial and body reference frames for the UAV, along with the direction of rotation. The body-frame x -axis is considered along the wingspan, the z -axis is considered perpendicularly upwards, and the y -axis completes the triad according to the right-hand rule. (B) Representation of aerodynamic forces on a single blade element.

where $\mathbf{p}^I = [p_x, p_y, p_z]^T$ is the position vector in the inertial frame, $\mathbf{R}_B^I$ represents the rotational transformation from body-frame to inertial-frame, $\mathbf{F}^B = [F_x, F_y, F_z]^T$ is the force vector, and $\mathbf{G} = [0, 0, g]^T$ is the gravity vector.

Similarly, the attitude dynamics equation of the UAV can be expressed as,

$$\mathbf{I}^B \dot{\boldsymbol{\omega}}^B + \boldsymbol{\omega}^B \times \mathbf{I}^B \boldsymbol{\omega}^B = \boldsymbol{\tau}^B \quad (6)$$

where $\mathbf{I}^B \in \mathbb{R}^{3 \times 3}$ is the inertia matrix, $\boldsymbol{\omega}^B$ is the angular velocity in the body frame, and $\boldsymbol{\tau}^B = [\tau_x, \tau_y, \tau_z]^T$ is the torque vector.

The aerodynamic forces acting on the wing are modeled with the blade element theory (BET) [3]. Under BET, the wing surface is split into n blade

elements, and the overall lift and drag contributions are obtained by summing up the individual contributions. The lift and drag force contribution from each blade element is calculated using,

$$dL = \frac{1}{2} C_l \rho U^2 c dr, \quad dD = \frac{1}{2} C_d \rho U^2 c dr \quad (7)$$

where C_l and C_d are the lift and drag coefficients respectively, ρ is the density of air, U is the relative air velocity encountered at the tip of the blade element, c is the chord length of each blade element, and dr is the width of each blade element. dL and dD can be resolved into normal and tangential forces as,

$$dN = dL \cos \theta + dD \sin \theta, \quad dT = dL \sin \theta - dD \cos \theta \quad (8)$$

where θ is the pitch angle between the relative air velocity and wing, as defined in figure 7(B).

3.2. Cyclic control

FROW's attitude is governed by a cyclic control, similar to the one onboard helicopters. Contrary to the swash plate mechanism, the motor(s) is used to provide a periodic thrust, which controls its roll and pitch. The cyclic control implemented is based on a square cyclic control strategy, as has been previously implemented in [19].

The amplitude of the cyclic command, T_{amp} , is mapped to the amplitude of the roll and pitch input, ϕ_c and θ_c respectively. T_{amp} and the direction control variable ψ_c of pitch and roll input are calculated as,

$$\psi_c = \text{atan2}\left(\frac{\phi_c}{\theta_c}\right), \quad T_{amp} = k_c \sqrt{\phi_c^2 + \theta_c^2} \quad (9)$$

where k_c is a constant to scale the effectiveness of roll and pitch actuation commands. The cyclic commanded thrust for the motor includes both the altitude correction (collective thrust) as well as direction correction (cyclic thrust), and is computed as,

$$T_{cyc} = \begin{cases} T_o + T_{amp}, & \text{if } \sin(\psi + \psi_c + \psi_0) > \epsilon \\ T_o - T_{amp}, & \text{otherwise} \end{cases} \quad (10)$$

where T_o is the thrust offset for maintaining altitude, ψ is the current azimuth heading, ψ_0 is the offset value for angular correction induced due to gyroscopic precession and other effects, and ϵ is the variable to control the duty cycle. In a manual flight, T_o is mapped directly to the throttle value from a radio controller. Likewise, the roll and pitch inputs required to calculate T_{amp} are mapped directly to the roll and pitch stick, respectively.

To conduct experiments in a motion capture environment, a closed-loop controller is designed. This consists of an attitude stabilizer control and a position controller.

3.3. Attitude control

FROW is affected by both aerodynamic forces and gyroscopic precession, similar to other rotating platform aerial vehicles. Similar to them, it has a natural precession circle that can either grow (unstable), remain constant (marginally stable), or decrease (stable) depending on the physical characteristics. Ideally, the physical dynamics of a monocopter should be set up so that the precession circle decreases without any control input. This ensures that when a disturbance occurs, the monocopter will fly in spirals with decreasing radius. Bai *et al* have presented one such passively stable revolving-wing drone in [7]. However, FROW-A's design leads to unsteady precession motion, and changing the wingspan mid-flight alters the moment of inertia, resulting in further

instability. Nevertheless, a proportional attitude stabilizer controller can regulate the precession motion to achieve stable flight [20].

The attitude stabilizer control command is generated as,

$$\mathbf{u}_{ac} = -\mathbf{k}_{ac} \cdot \boldsymbol{\Omega} \quad (11)$$

where $\mathbf{k}_{ac}$ is a positive gain matrix and $\boldsymbol{\Omega}$ is the angular velocity vector. Through this stabilization, the undesired angular velocities in the pitch and roll axis can be damped out. Since the attitude stabilization of the vehicle is desired in the roll and pitch axis only, $\mathbf{u}_{ac}[3]$ can be ignored. The effect of attitude stabilization has been further explained in the next section along with the experimental results.

3.4. Position control

The position controller of the platform is based on the sliding mode control (SMC) method. The sliding surface $\mathbf{s}$ has been defined using the error in position as well as velocity to keep control over both states,

$$\mathbf{s} = \mathbf{k}_p(\mathbf{p}_d - \mathbf{p}) - \mathbf{k}_v \mathbf{v} \quad (12)$$

where $\mathbf{k}_p$ and $\mathbf{k}_v$ are the sliding surface gains matrices for the position and velocity components, respectively.

The control law can be defined using standard SMC method [32] as,

$$\mathbf{u} = \mathbf{C} \cdot \text{sat}(\mathbf{s}) \quad (13)$$

where $\mathbf{C}$ is the gain matrix, and $\text{sat}(\cdot)$ is a saturation function defined as,

$$\text{sat}(\cdot) = \begin{cases} \text{sign}(\cdot), & \text{if } |\mathbf{s}| > s_\phi \\ \mathbf{s}/s_\phi, & \text{otherwise} \end{cases} \quad (14)$$

where s_ϕ is the boundary layer thickness of the sliding surface.

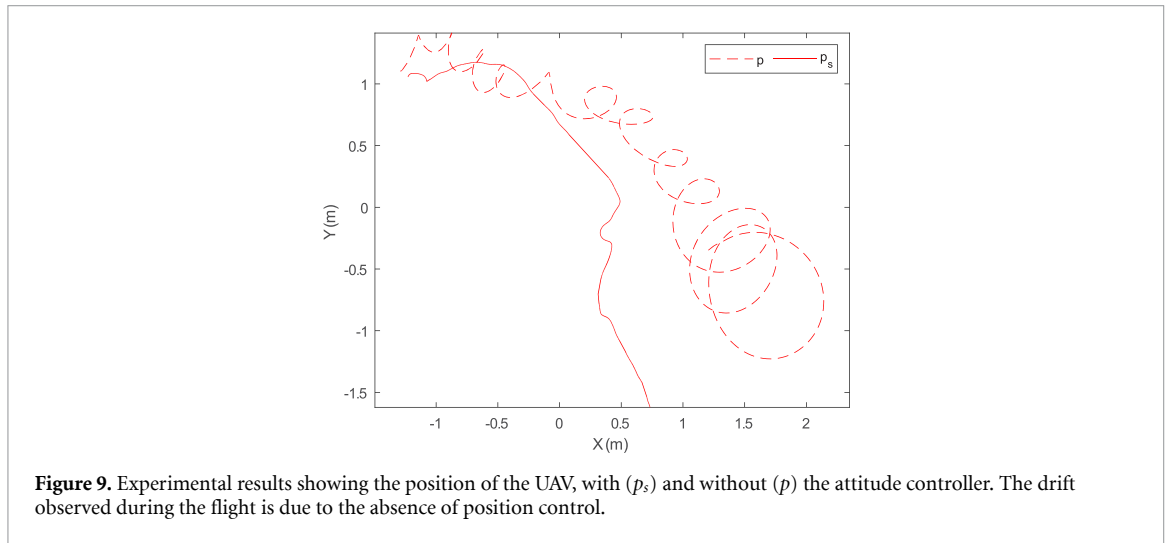
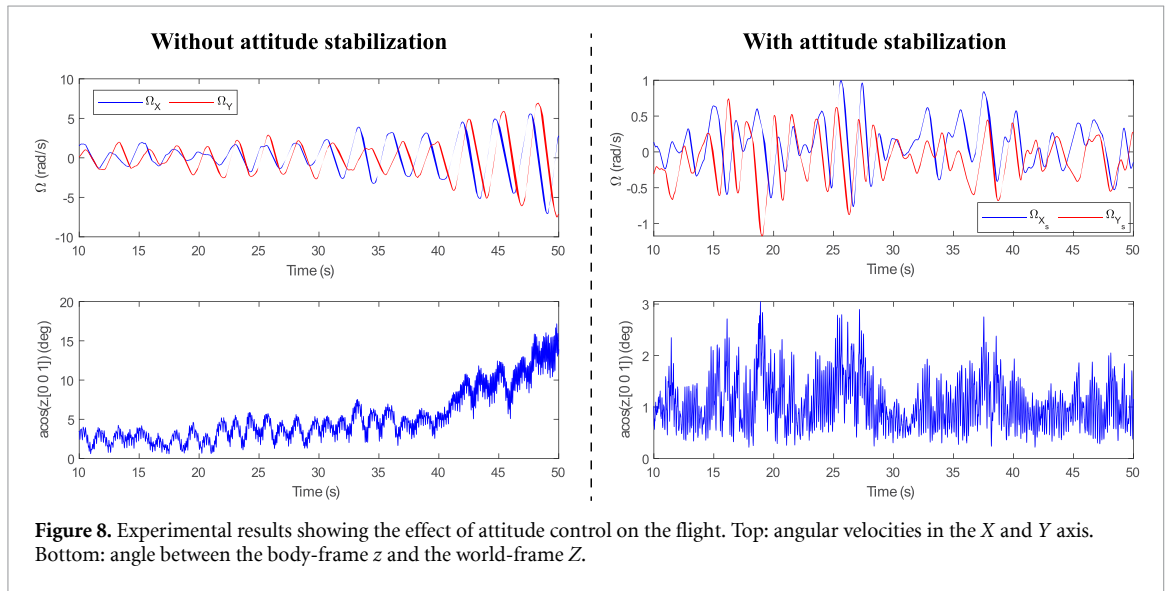
The final control input for roll and pitch in the cyclic controller are obtained by combining the controller outputs for roll and pitch axis from the attitude stabilizer control and position control [20] as,

$$\begin{bmatrix} \phi_c \\ \theta_c \end{bmatrix} = \begin{bmatrix} \mathbf{u}[2] + \mathbf{u}_{ac}[2] \\ \mathbf{u}[1] + \mathbf{u}_{ac}[1] \end{bmatrix} \quad (15)$$

and $T_o = \mathbf{u}[3]$. The obtained roll and pitch commands from equation (15) can be fed into equation (9) to obtain motion in desired axis.

4. Experimental results

In this section, we discuss the experimental setup used for conducting the indoor experiments for FROW-A along with the results obtained for closed loop waypoint and trajectory tracking in both extended as well



as folded configurations. Furthermore, we discuss the outdoor experiments conducted for FROW-P along with the results obtained.

The experiments are conducted in a motion-capture environment to verify the position control of the FROW. The motion-capture cameras capture the position and attitude of the reflective markers mounted on the FROW prototype. This position and attitude data is sent to a PC running the controller code, which computes the output commands for actuators. The output for motors and servos is sent through Wi-Fi to FROW. A second RC receiver under a human operator's control is also able to send control signals to FROW, which can overrule the output commands from the PC if desired.

In order to evaluate the impact of the attitude stabilizer on the flight performance, the UAV was commanded to hover at a designated altitude, both with and without the controller engaged. The obtained

results are presented in figure 8. During flights without attitude stabilization, it was observed that FROW-A experienced unstable motion, characterized by the increasing angular velocity of the X and Y axes, as illustrated in figure 8. A second plot was generated to represent the deviation of the tip path plane (TPP) relative to the world frame. This was achieved by plotting the angular difference between the body-frame z -axis and the world frame Z -axis. The use of attitude control resulted in a stable flight with no uncontrollable precession motion. FROW's position demonstrating the precession motion without the controller has been depicted in figure 9.

4.1. Closed-loop waypoint tracking

To test the closed-loop waypoint tracking of FROW-A, the UAV is tasked to fly toward waypoints forming a square shape of 2 m sides. Each waypoint is set for a 20 s interval. Figure 10 depicts the results

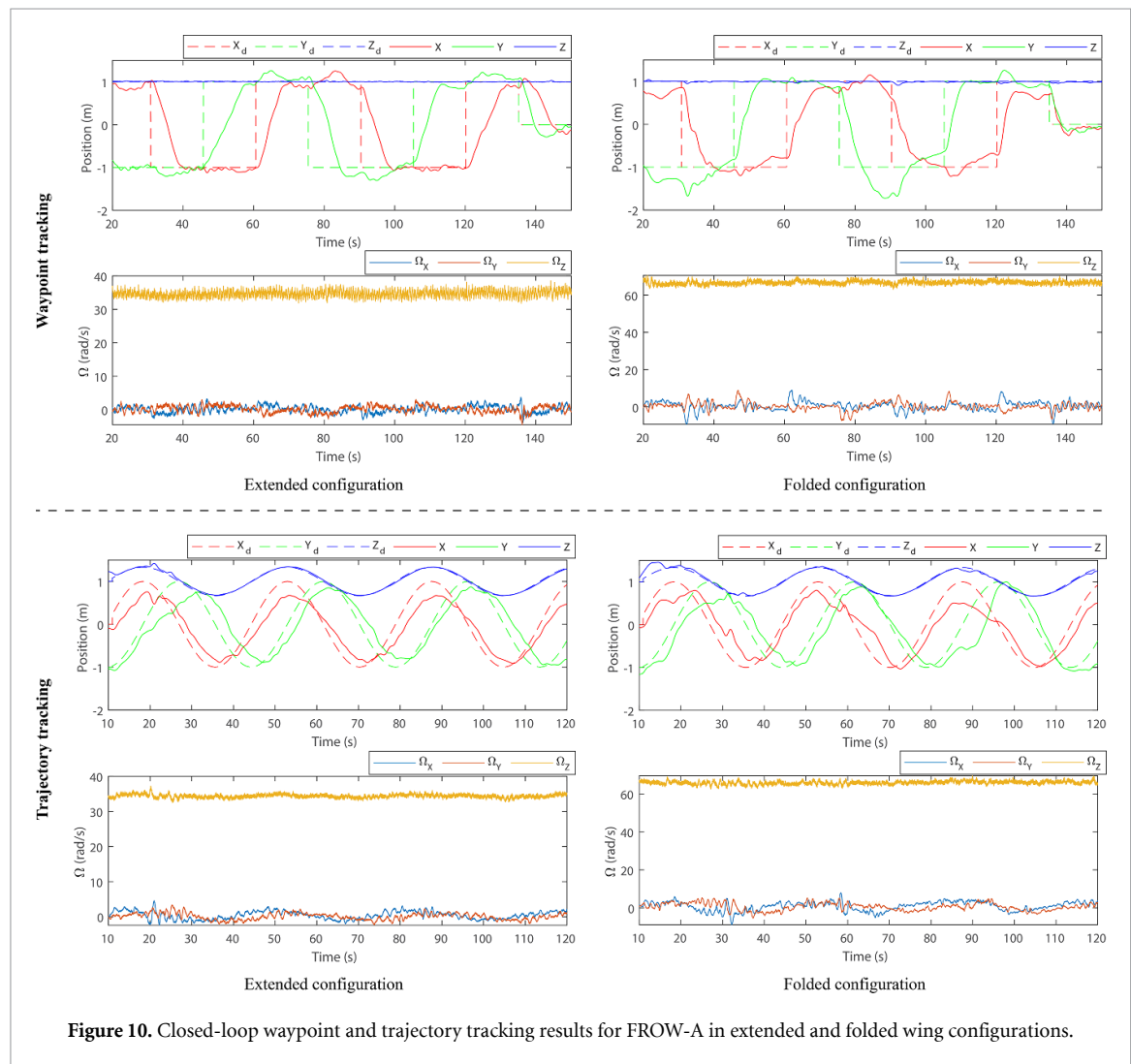


Figure 10. Closed-loop waypoint and trajectory tracking results for FROW-A in extended and folded wing configurations.

obtained for the extended and folded configurations. Alongside the position plots, Ω_x , Ω_y and Ω_z plot are also plotted. It was observed that both configurations performed well for the waypoint experiment, however, the folded configuration tends to overshoot the waypoint more compared to the extended one. This can be attributed to the higher angular momentum of the UAV in the folded configuration due to its higher Ω_z . Spikes in Ω_x and Ω_y can be observed in both cases where the waypoints have moved to a new position due to the change in the TPP of the UAV.

4.2. Closed-loop trajectory tracking

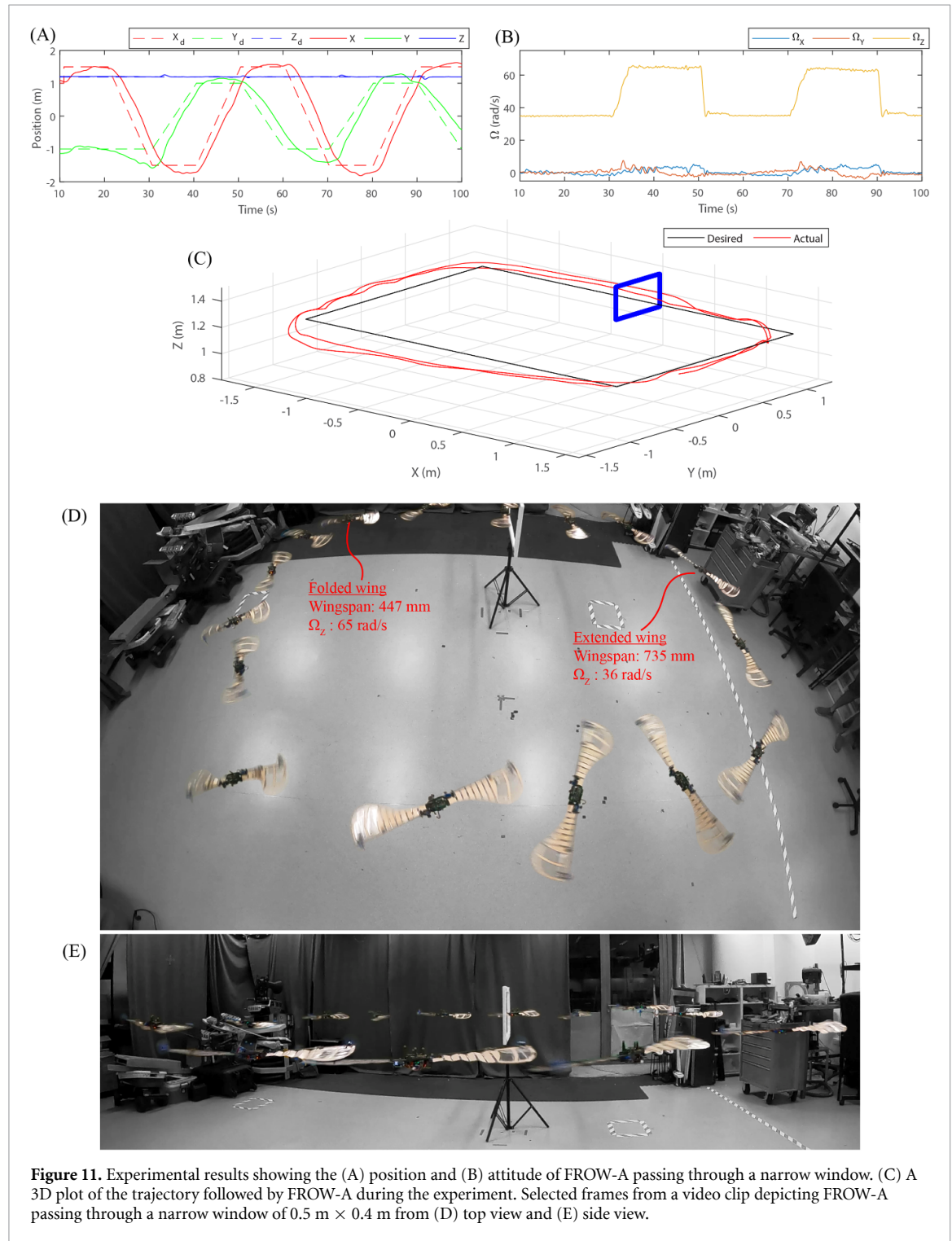
To test the closed-loop trajectory tracking of FROW-A, the UAV is tasked to fly in a helical trajectory, where the position in X, Y, and Z axes is changed continuously. A 1 m radius circle is traced in the X and Y axes, changing the desired position incrementally at each time step. Figure 10 depicts the results obtained for the extended and folded configurations. Overall, both configurations produced satisfactory results while tracking the trajectory. Similar

to waypoint tracking, a higher angular velocity in the Z axis can be observed for the folded configuration. In certain instances (for example around 30 s and 60 s), some peaks can be observed for the folded configuration, which can be associated with the controller trying to correct the position.

4.3. Flight test for contraction and expansion of the wing

To test the contraction and expansion mechanism of the wing during the flight, we performed several manual flight tests. The wing was commanded to fold and expand fully, and the flight characteristics were observed. It was observed that FROW-A takes about 3 s to fully fold and extend its wings at the maximum folding and unfolding speed achievable using the servos employed.

We performed experiments to test the mechanism's performance while tracking a square-shaped trajectory. To demonstrate the maneuverability of FROW-A, a rectangular cutout in the shape of a narrow window was placed along one of the edges of the



square. Figure 11 depicts the performance of the UAV during the experiment, meanwhile, figure 11 shows the frames extracted from the video where the FROW-A can be seen flying during the experiment. It was observed that while transitioning from the expanded to the folded configuration, the altitude of the prototype experiences some perturbations. This may be removed by further tuning of the controller and pulse width modulation (PWM) offset value of motors.

We tested FROW-A's performance by gradually changing the wingspan over a longer duration of time to ensure that the prototype is controllable at any length of wingspan possible with the mechanism. Figure 12 depicts the results obtained. A gradual increase and decrease in the rotation speed can be observed as the wings are folded in and expanded in a 15 s duration. For this experiment, the angular velocity increased from around 36 rad s⁻¹ at maximum

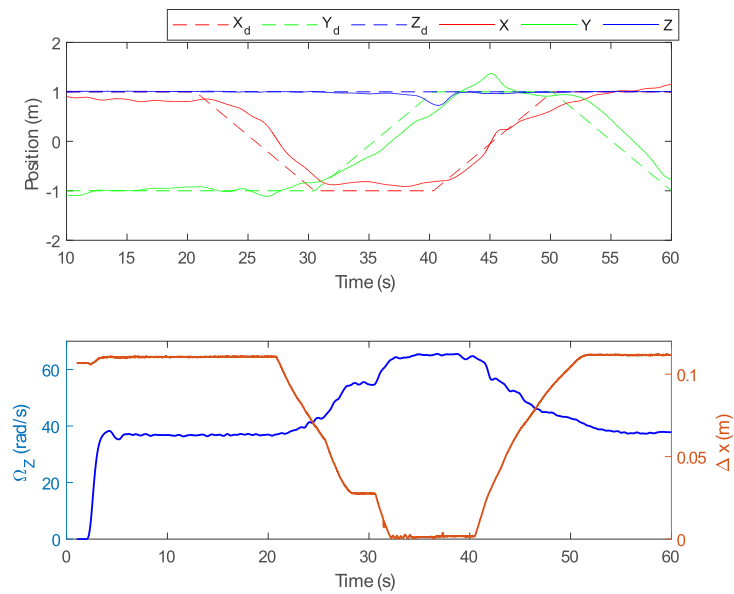


Figure 12. Experimental results showing the position control along with the angular velocity and displacement Δx of the wingspan of FROW-A.

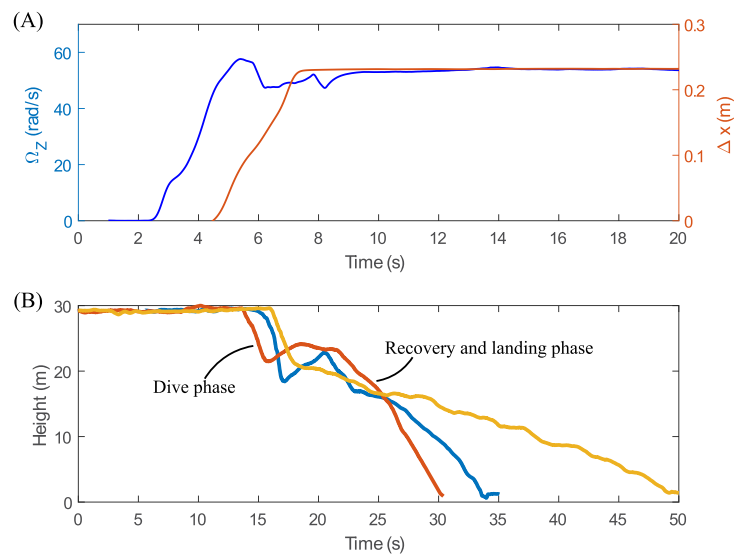


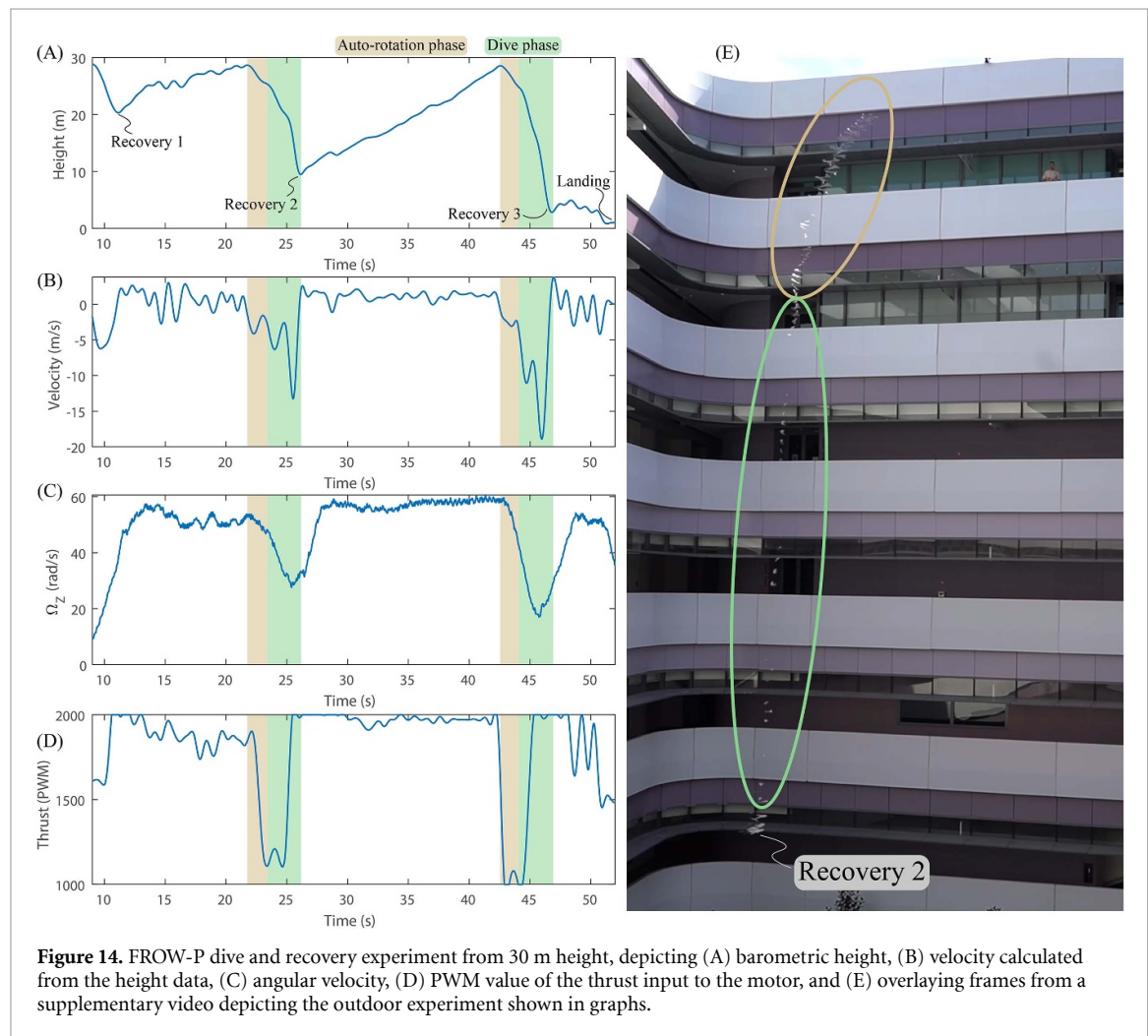
Figure 13. (A) Angular velocity of FROW-P during take-off and hovering. (B) Barometer data depicting the dive and recovery of FROW-P from 30 m altitude.

wingspan to around 65 rad s^{-1} at minimum wingspan. The position graph shows that the prototype successfully followed the trajectory during the experiment. To avoid extra current being drawn by the continuous servos, they are run at 80% of their power. However, once they reach their maximum performance limit at this level, the power is changed to 100%. This leads to a sudden change in the Δx , which can be observed around 30 s into the experiment.

Figure 13(A) depicts the performance of FROW-P during its take-off and hovering. It can be observed that the Ω_Z reaches about 58 rad s^{-1} on take-off to

overcome the spring force, gravity, and friction force. The plot also depicts the change in wingspan during the take-off. It can be observed that the wingspan slowly starts to increase after reaching around 43 rad s^{-1} . During hover, the Ω_Z is about 54 rad s^{-1} . Figure 5 depicts FROW-P taking off from stand still, where the expansion of the wing can be noticed with the increase in the angular velocity.

The spring parameters of FROW-P's folding mechanism were carefully chosen based on experimentation. Two key points were considered for the selection of these parameters. Firstly, there should



be enough force at zero Δx , so that the wing does not unfold itself. This ensures easier handling of the UAV for storage or while taking off. Secondly, in case we switch off the motor while flying, the prototype was designed to have a sufficient centrifugal force so that the wing does not automatically fold. This ensures that the wing does not fold and simply crash in the event of a power failure. The natural tendency of the platform to auto-rotate will help it gradually descend. We implemented a reverse thrust mechanism on the motor onboard FROW-P to rapidly decrease the angular velocity below the required amount, enabling it to dive.

Figure 13(B) depicts the results obtained for FROW-P in an outdoor experiment, where the prototype is hand thrown from a 30 m altitude to imitate a launch from a UAV. After passing through an initial dive phase for about 10 m, the ability to recover from a dive has been portrayed. Several experiments were performed and results for three different experiments have been depicted.

Figure 14 depicts the results obtained from the dive and recovery experiment performed. The experiment shows firstly the recovery of FROW-P after it is hand thrown from a height of 30 m. After flying back to nearly the same altitude, the reverse thrust mechanism is turned on to initiate the dive mode. Since the motors are rotating at high revolutions per minute (RPM), the reverse thrust mechanism takes a couple of seconds to turn on, during which FROW-P falls while auto-rotating. FROW-P is allowed to fall for about 20 m and is recovered again to repeat the experiment for an even deeper fall (25 m). The velocity for the prototype was obtained using differentiation, as a result, a lot of noise was obtained which was filtered to obtain the velocity profile. In a similar manner, the angular velocity profile is obtained from the heading angle data recorded from the magnetometer. During the first drop, the maximum velocity of the prototype is about -13 m s^{-1} , whereas, for the second drop, the maximum velocity is about -18 m s^{-1} . The experiment shows the ability of FROW-P to fully collapse its wing while in flight using the applied mechanism and

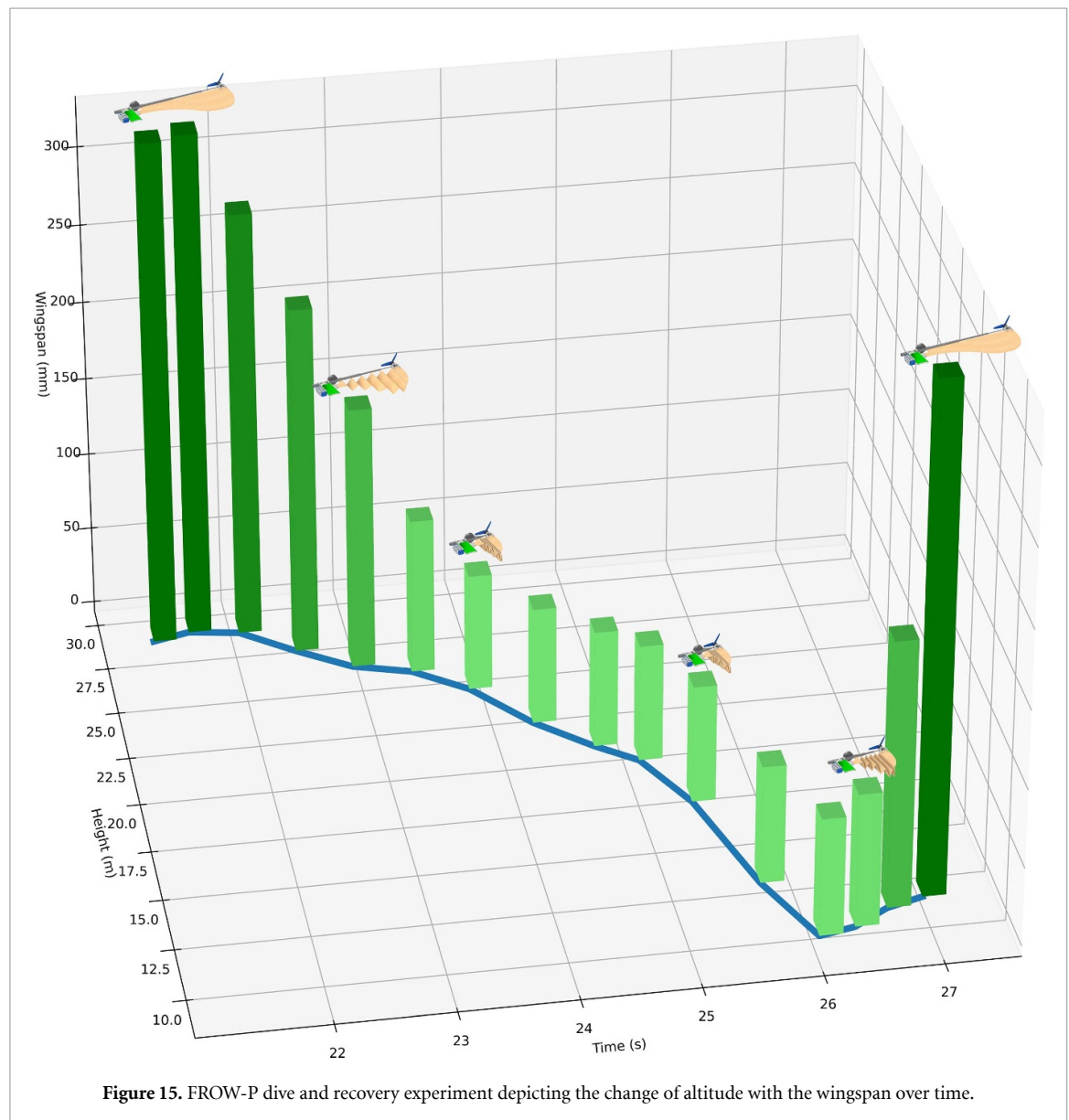


Figure 15. FROW-P dive and recovery experiment depicting the change of altitude with the wingspan over time.

recover at various distances while falling with different velocities. Figure 15 depicts the change of altitude and wingspan of FROW-P over time for recovery 2 shown in figures 14(A) and (E).

5. Conclusion

The concept of folding has been explored in various flying robots during flight, both in fixed wings as well as quadcopters through different implementations. Our UAV is inspired by the naturally occurring samara seeds, which have the capability of auto-rotating while falling, as well as inspiration has been drawn from different birds that can fold their wings and plunge toward prey or a water body. FROW is the first rotorcraft prototype, to have the ability to fold its wings, actively or passively, during flight.

We have presented results based on experimentation performed on the physical prototypes constructed through the design principles discussed in this

paper. The actively foldable concept presented can achieve stable flight and position control in both expanded and well as folded configurations. The stable flight achieved during the contacted wing configuration shows the ability of the UAV to navigate through narrow spaces, as has been presented in the supplementary video.

The passively foldable prototype presented in this paper is a novel monocopter configuration, that expands its wing under the influence of the centrifugal force generated while rotating. The spiral spring mechanism helps in retracting the wing, as soon as the centrifugal force is overpowered by the spring force. We presented the modeling of the spring and centrifugal forces for this prototype and showed the relationship between the rotational speed and the displacement of the wing. The model was further verified experimentally through recorded flight data and videos. We demonstrated the ability of the UAV to fold its wing and dive through the air and eventually

recover from the dive through the expansion of the wing.

Future works include enhancing the performance through optimization of wing shape as well as other flight characteristics.

Data availability statement

The data cannot be made publicly available upon publication because the cost of preparing, depositing and hosting the data would be prohibitive within the terms of this research project. The data that support the findings of this study are available upon reasonable request from the authors.

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References

- [1] Mahony R, Kumar V and Corke P 2012 Multirotor aerial vehicles: modeling, estimation and control of quadrotor *IEEE Robot. Autom. Mag.* **19** 20–32
- [2] Floreano D and Wood R J 2015 Science, technology and the future of small autonomous drones *Nature* **521** 460–6
- [3] Leishman J G 2006 *Principles of Helicopter Aerodynamics* (Cambridge: Cambridge University Press)
- [4] Driessens S and Pounds P 2015 The triangular quadrotor: a more efficient quadrotor configuration *IEEE Trans. Robot.* **31** 1517–26
- [5] Qin Y, Xu W, Lee A and Zhang F 2020 Gemini: a compact yet efficient bi-copter UAV for indoor applications *IEEE Robot. Autom. Lett.* **5** 3213–20
- [6] Chen N, Kong F, Xu W, Cai Y, Li H, He D, Qin Y and Zhang F 2023 A self-rotating, single-actuated UAV with extended sensor field of view for autonomous navigation *Sci. Robot.* **8** eade4538
- [7] Bai S, He Q and Chirarattananon P 2022 A bioinspired revolving-wing drone with passive attitude stability and efficient hovering flight *Sci. Robot.* **7** eabg5913
- [8] Cai X, Win S K H, Bhardwaj H and Foong S 2023 Modeling, control and implementation of adaptive reconfigurable rotary wings (AROWs) *IEEE/ASME Trans. Mechatronics* (<https://doi.org/10.1109/TMECH.2023.3235346>)
- [9] Houghton J and Hoburg W 2008 Fly-by-wire control of a monocopter *Project Report* (MIT)
- [10] Pounds P and Singh S 2015 Samara: biologically inspired self-deploying sensor networks *IEEE Potentials* **34** 10–14
- [11] Jameson S and Fregene K, Chang M, Allen N, Youngren H and Scroggins J 2012 Lockheed martin's samarai nano air vehicle: challenges research and realization *50th AIAA Aerospace Sciences Meeting* (1–21 January 2012)
- [12] Xu W, Cai Y, He D, Lin J and Zhang F 2022 FAST-LIO2: fast direct LiDAR-inertial odometry *IEEE Trans. Robot.* **38** 2053–73
- [13] Tan C H bin Shaiful D S, Tang E, Khaw J Y, Soh G S and Foong S 2020 Flydar: magnetometer-based high angular rate estimation during gyro saturation for SLAM *2020 IEEE Int. Conf. on Robotics and Automation (ICRA)* pp 8532–7
- [14] Win L S T, Win S K H, Sufiyan D, Soh G S and Foong S 2020 Achieving efficient controlled flight with a single actuator *2020 IEEE/ASME Int. Conf. on Advanced Intelligent Mechatronics (AIM)* pp 1625–31
- [15] Bhardwaj H, Cai X, Win S K H and Foong S 2022 Design, modeling and control of a two flight mode capable single wing rotorcraft with mid-air transition ability *IEEE Robot. Autom. Lett.* **7** 11720–7
- [16] Low J E, Win L S T, Shaiful D S B, Tan C H, Soh G S and Foong S 2017 Design and dynamic analysis of a transformable hovering rotorcraft (THOR) *2017 IEEE Int. Conf. on Robotics and Automation (ICRA)* pp 6389–639
- [17] Cai X, Win S K H, Win L S T, Sufiyan D and Foong S 2022 Cooperative modular single actuator monocothers capable of controlled passive separation *Proc. Int. Conf. on Robotics and Automation (ICRA)* pp 1989–95
- [18] Ulrich E R, Pines D J and Humbert J S 2010 From falling to flying: the path to powered flight of a robotic samara nano air vehicle *Bioinspir. Biomim.* **5** 045009
- [19] Bhardwaj H, Win S K H, Win L S T, Sufiyan D and Shaohui F 2021 P.I.D. Based sliding mode control of asynchronous multi-actuator monocothers *2021 IEEE/ASME Int. Conf. on Advanced Intelligent Mechatronics (AIM)* pp 239–46
- [20] Win S K H, Win L S T, Sufiyan D and Shaohui F 2021 Design and control of the first foldable single-actuator rotary wing micro aerial vehicle *Bioinspir. Biomim.* **16** 066019
- [21] Daler L, Mintchev S, Stefanini C and Floreano D 2015 A bioinspired multi-modal flying and walking robot *Bioinspir. Biomim.* **10** 016005
- [22] Tuna T, Ovrur S E, Gokbel E and Kumbasar T 2018 FOLLY: a self foldable and self deployable autonomous quadcopter *2018 6th Int. Conf. on Control Engineering and Information Technology (CEIT)* pp 1–6
- [23] Falanga D, Kleber K, Mintchev S, Floreano D and Scaramuzza D 2019 The foldable drone: a morphing quadrotor that can squeeze and fly *IEEE Robot. Autom. Lett.* **4** 209–16
- [24] E Ajanic Feroskhan M, Mintchev S, Noca F and Floreano D 2020 Bioinspired wing and tail morphing extends drone flight capabilities *Sci. Robot.* **5** eabc2897
- [25] Bouman A, Nadan P, Anderson M, Pastor D, Izraelevitz J, Burdick J and Kennedy B 2020 Design and autonomous stabilization of a ballistically-launched multirotor *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)* pp 8511–7
- [26] Mintchev S, Daler L, L'Eplattenier G, Saint-Raymond L and Floreano D 2015 Foldable and self-deployable pocket sized quadrotor *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)* pp 2190–5
- [27] Dufour L, Owen K, Mintchev S and Floreano D 2016 A drone with insect-inspired folding wings *2016 IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)* pp 1576–81
- [28] Bai S, Ding R and Chirarattananon P 2022 A micro aircraft with passive variable-sweep wings *IEEE Robot. Autom. Lett.* **7** 4016–23
- [29] Bucki N, Tang J and Mueller M W 2023 Design and control of a mid air reconfigurable quadcopter using unactuated hinges *IEEE Trans. Robot.* **39** 539–57
- [30] Win S K H, Win L S T, Sufiyan D, Soh G S and Foong S 2022 An agile samara-inspired single-actuator aerial robot capable of autorotation and diving *IEEE Trans. Robot.* **38** 1033–46
- [31] Krishna Rao T 2013 *Design of Machine Elements* vol 2 (New Delhi: IK International Pvt Ltd)
- [32] Gao W and Hung J C 1993 Variable structure control of nonlinear systems: a new approach *IEEE Trans. Ind. Electron.* **40** 45–55